Pooled Accuracy Is Not Enough to Authorize VMD Preprocessing: Same-Source and External Bearing Audits

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Abstract

Pooled diagnostic accuracy is widely used to decide whether preprocessing should be enabled, although measurement condition, validation unit, and class mix can alter what that number means. We audit this authorization problem using same-source and external bearing evidence. On HUSTbearing, fixed K3 shows a clean-trained speed-level gain of 9.72 pp, but matched 0 dB noise augmentation on both raw and K3 reverses the effect to -7.94 pp (unit-only interval [-12.61, -3.44]; crossed robustness interval [-12.72, -3.28]) and returns Not qualified under the illustrative $\delta = 1$ pp margin. The bearing-level effect is 0.49 pp [-0.62, 1.53], and bearing-wise pooled balanced accuracy stays below five-class chance. The speed-minus-bearing contrast of 10.21 pp [3.67, 19.06] is descriptive across different unit sets, not a paired per-bearing interaction. In an independent PHM2012 run-to-failure audit with 17 non-fragile bearings, clean-trained K3 raises mean balanced accuracy by 24.51 pp, but the healthy class rises by 70.75 pp while the degraded class falls by -21.73 pp ([-43.14, -5.39]); class safety therefore fails. Matched augmentation changes the mean effect to -6.29 pp. No external block receives a full Qualified status. A three-state evidence audit distinguishes Not qualified from Insufficient and organizes the decision, but it is a decision aid, not a validated certification framework. The empirical contribution is that pooled gains can reverse under matched measurement conditions and conceal class-specific harm. Raw remains the conservative default.

Keywords: rolling bearing diagnosis; measurement-conditioned preprocessing; pooled accuracy; validation unit; class-specific harm; matched noise augmentation

1 Introduction

Bearing faults are a common cause of degradation in rotating machinery, and vibration diagnosis is most useful when fault evidence can be separated from competing components [1]. Variational mode decomposition (VMD) has become a frequent preprocessing choice because it decomposes a non-stationary signal into a small bank of band-limited modes [2]. Studies have applied VMD to rub-impact diagnosis, wheel-set bearings, incipient faults, locomotive bearings, and noise-robust diagnosis more generally [3–7]. Reviews of intelligent diagnosis reinforce the same motivation: better representations can make downstream classification easier [8].

The practical question is not whether VMD can improve a carefully selected example. It is whether a reported gain constitutes sufficient evidence to enable decomposition for a new measurement. Three research streams are relevant but do not answer that authorization question. The first optimizes VMD parameters within one benchmark family [3, 4, 6, 9–13]. It finds a good local decomposition but does not establish whether that decomposition should be enabled for a different sensor, bearing geometry, load profile, or class distribution. The second evaluates whether diagnostic representations remain useful under changing domains and operating conditions [14, 15].

The third studies data and evaluation practice, including leakage, benchmark reuse, and optimistic transfer [16]. These streams provide methods and diagnostics, but they do not determine what a reported gain authorizes.

We audit two ways in which that authorization can fail. First, the same HUSTbearing source is evaluated under a clean-trained baseline and under matched noise augmentation. A clean-trained speed-level gain of 9.72 pp reverses to -7.94 pp once both representations are trained under the same 0 dB measurement condition. The same-source speed and bearing holdouts also give different evidence scopes, although their contrast is descriptive rather than a paired per-bearing interaction. Second, an independent PHM2012 run-to-failure audit with 17 non-fragile bearings is used as a locked external block. There, clean-trained K3 improves mean balanced accuracy by 24.51 pp, but the healthy class improves by 70.75 pp while the degraded class falls by -21.73 pp. The apparent pooled gain therefore coincides with class-specific harm.

The validation unit is the entity held out and generalized over: a motor speed, a physical bearing, a fault-size group, or a factory case. An evidence block is a declared family of completed measurements sharing a rig, protocol, or operating role. The two terms are not interchangeable. Evidence blocks organize the audit; validation units determine what inference each block can authorize. A gain observed under an operating-speed holdout does not establish a gain under physical-bearing holdout, and a clean-versus-noise distinction does not determine the sign of the downstream effect.

We use a three-state evidence audit to organize the results. Paired unit-level effect and class-damage intervals are compared with application-defined margins and mapped to Qualified, Not qualified, or Insufficient. Missing or fragile class-level evidence returns Insufficient rather than a weaker pass. CCMG is retained only as an optional measurement-condition prefilter; it is not a benefit predictor and is not a main contribution.

We address three questions. First, does a clean-trained VMD advantage survive matched measurement conditions? Second, can a pooled improvement coexist with class-specific harm under an external target-unit holdout? Third, can a three-state audit make the difference between demonstrated failure and insufficient evidence explicit without treating the audit as a validated certification system?

The contribution is deliberately narrower than a new decomposition algorithm. It has three parts.

1. An empirical falsification of the clean-trained baseline: fixed K3 changes sign under matched noise augmentation, so the apparent advantage is not a sufficient measurement warrant.

2. An external class-level counterexample: the 17-bearing PHM2012 audit shows a large pooled gain together with an entirely negative degraded-class interval.

3. A three-state evidence audit that makes the authorization boundary explicit. It is a decision template applied to established intervals, not a validated qualification framework.

For measurement practice, the contribution is the empirical demonstration that pooled accuracy can misauthorize preprocessing and the decision vocabulary that prevents that result from being read as deployment evidence. The paper does not propose a new VMD optimizer, classifier, sensor-calibration model, or uncertainty budget.

2 Literature Review

2.1 VMD parameter selection

VMD decomposes a non-stationary signal into K band-limited modes through a variational optimization procedure [2]. Its use in rotating-machinery diagnosis has motivated methods for selecting decomposition parameters and identifying modes that preserve diagnostically useful information [3, 4, 6, 9–13]. These methods optimize a criterion such as mode independence, envelope concentration, entropy, or fault-band energy on a development set and report performance on a test set drawn from the same benchmark family. This design is appropriate for finding a good local decomposition. It does not establish whether the selected operation should be enabled for a different sensor, bearing geometry, load profile, or class distribution. A criterion that ranks candidates under one protocol need not rank them under another.

2.2 Cross-domain evaluation and leakage-controlled practice

A second body of work asks whether diagnostic representations remain useful under changing domains and operating conditions [14, 15]. A third studies data and evaluation practice, including leakage, benchmark reuse, and optimistic transfer [16]. These streams establish that a reported number can rest on conditions the number itself does not carry. They provide diagnostics for whether an evaluation is valid; they do not specify what a valid evaluation authorizes for another declared unit. This distinction matters because a decomposition can increase spectral concentration while reducing discriminative information. It can suppress a broadband disturbance, merge nearby fault components, create correlated mode inputs, or change the relative difficulty of fault classes. A pooled gain can therefore coexist with substantial bearing-level reversals.

2.3 Intervals, equivalence margins, and evidence status

The individual statistical ingredients used here are established rather than new. Cross-validated model selection can produce optimistic error estimates [17], and inference after selection must account for the selection event [18]. Non-inferiority and equivalence studies formalize materiality and harm margins [19], while clustered or repeated-measure evaluations require the dependence structure to be respected [20]. These tools provide intervals and tests; they do not by themselves map validation-unit evidence to a decision. What is missing is an explicit mapping from paired validation-unit evidence to a decision that separates authorization, demonstrated failure, and insufficient evidence. A confidence interval that crosses the materiality margin does not prove failure; it fails to license release. A missing class-level interval is not a pass; it is absent evidence.

2.4 Positioning

The VMD parameter-selection literature supplies candidate decompositions. The cross-domain and evaluation-practice literature supplies leakage-aware splits and transfer diagnostics. The non-inferiority and equivalence literature supplies intervals and margins. This paper's empirical question is narrower: when does a reported pooled gain survive a change in measurement condition, validation unit, or class-level inspection? The three-state audit is used to organize that question and to prevent an unresolved interval from being read as a positive result. It is not presented as a new statistical test.

Table 1: Principal symbols used in Section 3.

Symbol	Description	Unit
Δ_u	Paired K3-minus-raw balanced-accuracy effect for unit u	pp
$[L_u, U_u]$	Two-sided unit-only interval for Δ_u	pp
δ	Minimum material benefit	pp
$r_{u,c}$	Paired recall change for class c in unit u	pp
$[L_{u,c}, U_{u,c}]$	Simultaneous class-level interval	pp
η	Maximum tolerated recall loss for any class	pp
s	CCMG composite score	dimensionless
θ	CCMG threshold	dimensionless

3 Model, Methodology, and Data

Table 1 summarizes the principal notation used in this section.

3.1 VMD representations and evaluation units

For a measured window $x(t)$, VMD estimates K modes $u_k(t)$ and center frequencies ω_k by solving

$$\min_{\{u_k\},\{\omega_k\}} \sum_{k=1}^{K} \left\| \partial_t \left[\left(\delta(t) + \frac{j}{\pi t} \right) * u_k(t) \right] e^{-j\omega_k t} \right\|_2^2, \tag{1}$$

subject to $\sum_k u_k = x$. The implementation fixes $\alpha = 2000$, $\tau = 0$, and convergence tolerance 10^{-6} across all compared representations. These are analysis constants, not evidence that $\alpha = 2000$ is universally optimal.

Raw samples are treated as a one-channel input; the K modes are treated as a multi-channel input. Within each external dataset, raw and VMD inputs use the same layer widths, optimizer, learning rate, batch size, and training schedule. The comparison therefore tests the complete representation pipeline under a shared training budget, not two capacity-matched encoders. The raw3 capacity control is post hoc and exploratory; its results are reported in Appendix Table A1 and should not be interpreted as a capacity-matched test.

The validation unit is the entity held out and generalized over. Windows, recordings, and repeated measurements within one unit are not independent validation units. An evidence block is a declared family of completed measurements sharing a rig, protocol, or operating role; its validation unit determines what inference the block can authorize.

3.2 Matched noise-augmentation protocol

All external datasets use 2,048-sample, non-overlapping windows normalized independently by their own mean and standard deviation. The matched noise-augmentation protocol described below applies specifically to the same-source HUSTbearing experiment in Sections 4.2 and 4.3. The remaining external blocks (HSE, HUST-Hanoi, SCA, and Mehran-UET) retain their original clean-trained evaluation protocols.

For the HUSTbearing experiment, the training/test split is fixed first, and noise is then added independently to training and test windows.

Raw branch. The raw branch receives the clean training windows plus a noisy copy of the same training windows, as a one-channel input. Test windows receive an independent noise realization.

K3 branch. The K3 branch receives the same clean training windows plus the identical noisy training copy, decomposed into three modes. Test windows receive the identical test noise realization used by the raw branch, then decomposed into three modes.

Both branches therefore share the same raw samples, labels, sample budget, classifier seed, training schedule, and noise realization. The paired K3-minus-raw difference is computed on the same held-out unit under the same noise draw.

Noise-realization structure. The matched-augmentation control is identified by the frozen split, training seed, and a single fixed 0 dB training/test realization set. The crossed interval resamples training seeds; noise-realization sensitivity is assessed separately in the clean-trained multi-realization analysis.

Noise-mismatch check. To test whether the matched-augmentation result depends on exact training/test noise matching, both representations are additionally trained at 0 dB and tested at -3 dB and $+4$ dB. These mismatch conditions are robustness checks, not the primary comparator.

3.3 PHM2012 external run-to-failure audit

The external audit uses the complete PHM2012/PRONOSTIA bearing lives [21]. Seventeen bearings are taken from the learning and full test sets. For each bearing, the first 20% of its life is labeled healthy and the last 20% is labeled degraded; the middle 60% is excluded. Twelve acquisitions per class per bearing are sampled at evenly spaced positions inside each interval. The horizontal acceleration channel is normalized per 2,048 sample window, and the physical bearing is the validation unit in a leave-one-bearing-out split. The audit uses three classifier seeds and compares raw with fixed K3 under the same clean-trained and matched-augmentation protocols. A single fixed 0 dB noise realization is shared by raw and K3; the result is therefore a locked external audit, not a multi-realization field study. This endpoint is binary healthy-versus-degraded run-to-failure classification and is not a reproduction of the five-class HUSTbearing fault type task.

3.4 Clean-condition measurement gate

CCMG is an optional prefilter, not a benefit predictor. For a target rig, let m_j and M_j be the median and median absolute deviation of feature j in a clean commissioning baseline. The normalized change is

$$z_j = \begin{cases} \dfrac{f_j - m_j}{\max(1.4826 M_j, 10^{-6})}, & j \in \mathcal{N}_{\uparrow}, \\ \dfrac{m_j - f_j}{\max(1.4826 M_j, 10^{-6})}, & j = \text{SNR proxy}, \end{cases} \tag{2}$$

where $\mathcal{N}_{\uparrow}$ contains spectral flatness, spectral entropy, high-frequency ratio, envelope entropy, and wavelet noise-to-RMS ratio. The composite score is

$$s = \frac{1}{|\mathcal{J}|} \sum_{j \in \mathcal{J}} z_j, \tag{3}$$

with $\mathcal{J}$ containing the noise-increase features and the SNR proxy. The threshold θ is selected only from rigs other than the held-out rig. Candidate thresholds are quantiles of the training scores, and the objective is balanced accuracy for clean versus any controlled-noise condition. The resulting decision answers a narrow measurement question: whether the signal has moved away from the

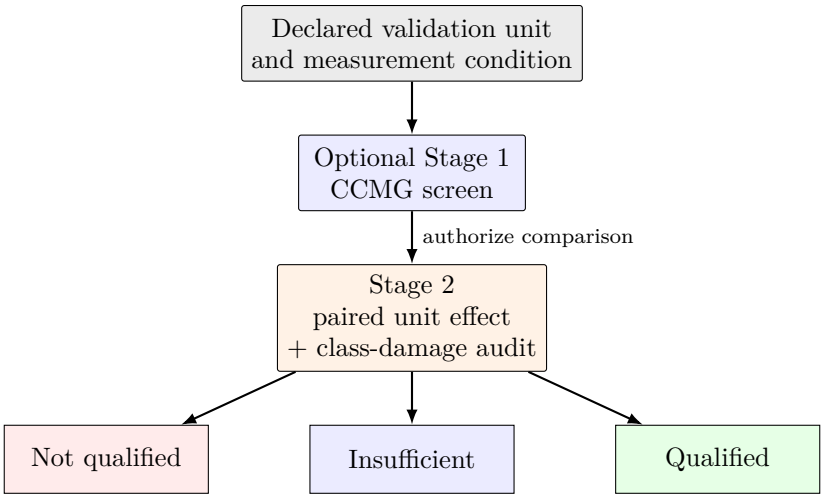


Figure 1: Three-state evidence audit used in this study. Stage 1 is an optional measurement-condition screen and may be omitted. Missing or fragile class-level evidence returns Insufficient. The audit is a decision aid, not a validated certification framework.

clean commissioning condition. It does not decide whether decomposition improves a downstream diagnosis.

3.5 Three-state evidence audit

Figure 1 summarizes the decision path from a declared validation unit to the three evidence states. The figure organizes the audit; it is not presented as a validated decision product.

We distinguish measurement-condition screening from validation-unit evidence status. For a declared unit u, let Δ_u be the paired K3-minus-raw balanced-accuracy effect and $[L_u, U_u]$ its two-sided $1-\alpha$ unit-only interval. Let $\delta \geq 0$ be the minimum material benefit and $\eta \geq 0$ the maximum tolerated recall loss for any class. For class c, let $r_{u,c}$ be the paired recall change and $[L_{u,c}, U_{u,c}]$ a simultaneous interval across classes at the declared confidence level. If class-level intervals are unavailable, the class-damage condition is treated as missing, not as satisfied.

The decision template is:

1. **Optional Stage 1: measurement-condition screening.** Compute s. If $s < \theta$, return raw as default and make no benefit claim. If $s \geq \theta$, authorize only a Stage 2 comparison. Stage 1 may be omitted when a clean commissioning baseline is unavailable or when measurement-condition screening is not required.

2. **Stage 2: validation-unit-specific evidence audit.** Estimate $[L_u, U_u]$ using paired folds that hold out the declared unit and check the worst class-level recall change.

3. Return **Qualified** only when $L_u > \delta$, all required class intervals are available, valid, and non-fragile, and $\min_c L_{u,c} \geq -\eta$. Return **Not qualified** when $U_u \leq \delta$ or $\exists c : U_{u,c} < -\eta$. Otherwise return **Insufficient**.

Class intervals are simultaneous over classes using Bonferroni adjustment at the declared confidence level. Bonferroni controls family-wise coverage across classes; it does not stabilize intervals estimated from few independent units. If a class has too few independent validation units to construct a valid interval, the evidence is recorded as missing. If the interval can be computed but

rests on two to five independent units, it is marked fragile and does not by itself license a Qualified decision. If the class-damage interval clearly exceeds η and the interval method is valid, the block can return Not qualified on class-harm grounds.

The current application is a retrospective worked example; it does not satisfy the prospective predeclaration condition. The non-fragile requirement is an audit requirement of this worked example, not a general small-sample statistical law.

3.6 Statistics and aggregation

We report paired percentage-point differences in test accuracy. Three statistical objects are kept separate:

Object	Statistical meaning	Use
Unit-only interval	Uncertainty across declared validation units, conditional on the evaluated protocol	Stage 2 decision interval
Crossed interval	Sensitivity to training-seed and, where available, noise-realization variation	Robustness check
Speed-minus-bearing contrast	Descriptive contrast between different held-out unit sets	Not a paired per-bearing interaction

For the discovery network we fit

$$\Delta_i = \mu + u_{\mathrm{rig}(i)} + v_{\mathrm{protocol}(i)} + \varepsilon_i, \tag{4}$$

where u_{rig} and v_{protocol} are independent mean-zero random effects.

To audit how conclusions depend on the validation unit, we also analyse 86 heterogeneous paired K3-minus-raw validation units nested in ten evidence blocks:

$$\Delta_{ij} = \mu + u_i + \varepsilon_{ij}, \tag{5}$$

where i indexes evidence blocks and j indexes units within a block. We report a moment-based variance decomposition, the corresponding intraclass correlation, and the range of pooled means obtained by leaving out one complete evidence block. The blocks are not exchangeable independent samples.

For the external physical-bearing audits, a held-out bearing receives all of its recordings under different speeds or loads. We report the mean and median paired effect across bearings, a bearing bootstrap interval, and the number of improved, unchanged, and worsened bearings. Pooled out-of-fold balanced accuracy is computed from summed confusion matrices and reported descriptively.

For the same-source HUSTbearing split, uncertainty is reported using 20,000 percentile-bootstrap draws. The unit-only interval resamples held-out speeds or bearings and is the Stage 2 decision interval. The crossed interval independently resamples held-out units and training seeds, and, where multiple noise realizations exist, noise realizations. It is reported as a robustness check.

4 Results and Discussion

4.1 A pooled mean does not define what a validation unit authorizes

Figure 2 summarizes how the held-out entity determines the scope of the authorized inference.

A pooled mean across 86 heterogeneous validation units in ten evidence blocks is 0.88 pp. Read in isolation, that positive value suggests that K3 helps. Table 2 shows why the pooled value is not

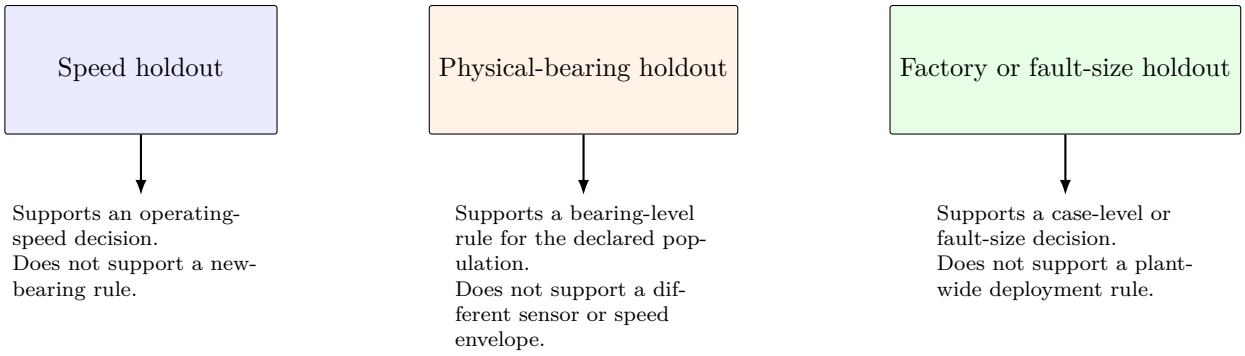


Figure 2: Validation units and the scope of inference. The held-out entity determines what a completed evaluation can authorize.

an authorization: the block means reverse in sign, and leaving out one complete evidence block moves the pooled mean between -0.58 pp and 2.93 pp. The same treatment can be positive for one block and negative for another because the blocks answer different questions. A motor-speed fold asks whether an operating-speed decision transfers. A physical-bearing fold asks whether the decision survives when the bearing itself changes. A fault-size fold asks how the class boundary responds. These are not interchangeable samples from one evidence block.

Read row by row, the table separates four kinds of authorization. CWRU, JNU, and MFPT establish controlled or rig-level sensitivity, not transfer to new bearings. Ottawa and Paderborn show that protocol composition can reverse the direction inside the discovery panel. HUSTbearing provides a clean-trained speed-level comparator, but the matched-augmentation audit does not authorize an operational decision. HSE, HUST-Hanoi, SCA, and Mehran-UET test four different physical or industrial units; their signs do not form one exchangeable sample.

The moment-based between-block standard deviation is 10.53 pp and the within-block standard deviation is 32.59 pp; the resulting ICC is 0.095. A maximum-likelihood fit places the between-block component near zero. These estimates do not support a claim that between-block variance dominates. They do not, however, make the validation unit irrelevant. Small between-block variance can coexist with sign reversals at the level of the declared unit, and block composition still changes the direction and scope of the pooled conclusion.

4.2 Same-source split: speed-wise versus bearing-wise

Figure 3 and Table 3 report the decisive same-source HUSTbearing split. The analysis holds the data source fixed while changing the validation-unit set. The unit-only interval is computed by aggregating the prespecified training seeds and noise realizations within each held-out unit, then resampling the independent held-out units. The 15 seed-by-noise aggregates are repeated experimental combinations, not 15 independent validation units. The single-realization speed-wise estimate is 9.72 pp with unit-only interval [6.83, 13.00]. The 15-aggregate speed-wise estimate is 11.00 pp with unit-only interval [7.88, 13.51] and crossed interval [6.72, 15.10]. The bearing-wise estimates are 0.21 pp (single realization, unit-only [-6.39, 5.56]) and 0.79 pp (15 aggregate, unit-only [-4.22, 5.78], crossed [-6.68, 5.72]). The speed-minus-bearing contrast is 10.21 pp (crossed interval [3.67, 19.06]) and corresponds to the 15-aggregate estimates; the single-realization contrast is 9.51 pp. The contrast compares different validation-unit sets and is not a paired per-bearing interaction.

The absolute pooled balanced accuracies also require interpretation. Speed-wise pooled out-

Table 2: Validation units and evidence boundaries. Mean effects are relative to clean-trained raw. The rows are evidence blocks, not exchangeable independent samples. The HUSTbearing value is the original speed-holdout run; Table 3 reports the separate same-source fixed-budget rerun.

Evidence block	Role	Unit	n	Mean Δ (pp)	Signs (+/0/−)	Supports	Does not support
CWRU	discovery rig	condition/file fold	1	+6.00	1/0/0	Controlled noise sensitivity	Cross-bearing transfer
JNU	discovery rig	condition fold	1	+25.50	1/0/0	Severe-noise sensitivity	Evidence-block-wide rule
MFPT	discovery rig	rig protocol	1	+44.00	1/0/0	Independent-rig response	Bearing-level generalization
Ottawa	discovery condition	condition fold	4	+0.59	2/0/2	Condition-wise reversals	Stable clean/noise boundary
Paderborn	discovery protocol	protocol fold	3	-0.31	1/0/2	Protocol sensitivity	Transferable mode-count rule
HUSTbearing	speed holdout	motor speed	10	+7.61	8/0/2	Operational speed holdout	Physical-bearing transfer
HSE	bearing holdout	physical bearing	11	+10.81	6/3/2	Leave-one-bearing-out test	Stable positive block effect
HUST-Hanoi	bearing holdout	physical bearing	33	-0.42	11/17/5	Leave-one-bearing-out test	General K3 benefit
SCA	industrial holdout	factory case	10	-4.11	5/0/5	Case-level transfer audit	Factory-wide deployment
Mehran-UET	fault-size holdout	fault-size group	12	-11.81	5/1/6	Decision-boundary audit	Fault-physics mechanism

Table 3: Same-source HUSTbearing split comparison and noise-realization sensitivity. Fixed rows use the original realization. The multi-realization rows aggregate three seeds and five noise realizations. Unit-only intervals are the Stage 2 decision intervals; crossed intervals are robustness checks.

Scheme	Realization	Units	Mean Δ	Unit-only 95% CI	Crossed 95% CI	Signs	Raw/K3 BA
Speed	fixed	10	+9.72	[6.83, 13.00]	–	10/0/0	22.44/32.17
Speed	15 aggregate	10	+11.00	[7.88, 13.51]	[6.72, 15.10]	9/0/1	22.94/33.94
Bearing	fixed	8	+0.21	[-6.39, 5.56]	–	5/2/1	3.40/3.61
Bearing	15 aggregate	8	+0.79	[-4.22, 5.78]	[-6.68, 5.72]	5/2/1	2.90/3.69

of-fold balanced accuracy is 22.44 % for raw and 32.17 % for K3. Bearing-wise pooled out-of-fold balanced accuracy is 3.40 % and 3.61 % under clean training, and 1.32 % and 1.81 % under matched augmentation, for a five-class problem whose uniform-random reference is 20 % (Table 4). The class-supported analysis excludes bearings whose training partition lacks at least one class; the eight retained bearings cover all five classes. Each physical bearing contributes one class, so the bearing-level value is a class-recall comparison, not a complete multi-class diagnostic capability estimate.

4.3 Matched noise augmentation reverses the clean-trained advantage

Figure 4 and Table 5 report the matched-augmentation and signal controls. The clean-trained comparison is deliberately vulnerable: clean-to-clean controls work (raw 77.56%, K3 72.56%), but clean-to-0 dB performance collapses (raw 22.44%, K3 32.17%). Noise-augmented raw reaches 63.83%, while band-pass and wavelet controls reach 41.83% and 22.00%. The clean-to-0 dB degradation is consistent with substantial train/test measurement shift. The band-pass and wavelet controls are trained on clean windows and tested at 0 dB; they are descriptive signal-processing controls, not matched-augmentation competitors.

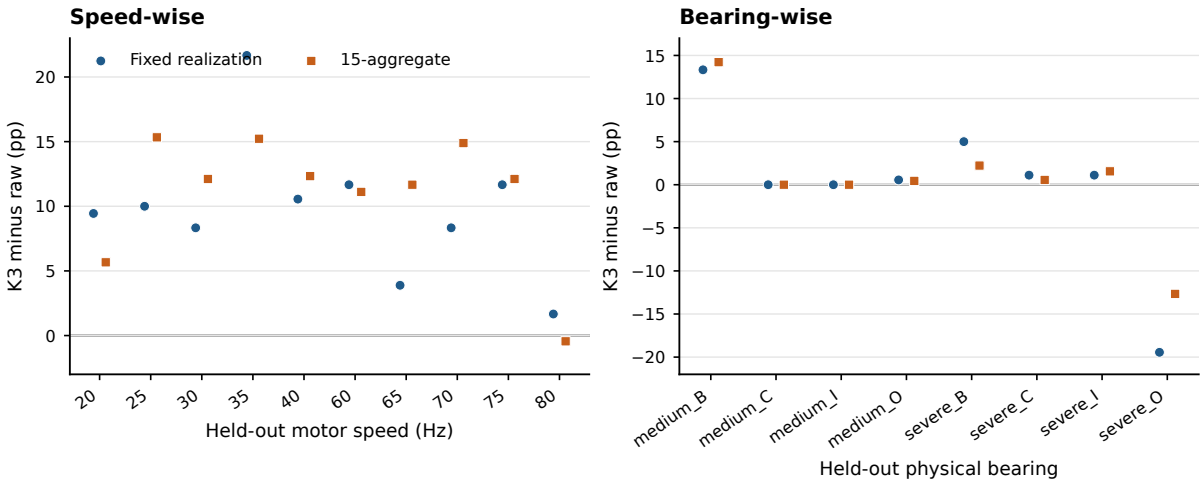


Figure 3: Same-source HUSTbearing split under the original realization and the 15 seed/noise aggregate. Each point is a held-out validation unit. The speed-level gain and bearing-level neutrality describe different estimands.

Table 4: Bearing-wise pooled out-of-fold balanced accuracy for the same eight class-supported bearings. BA $= (1/5)\sum_c \text{Recall}_c$. The 20% five-class uniform-random level is a reference baseline, not an operational capability threshold.

Training protocol	Validation scope	Raw BA	K3 BA
Clean-trained	Bearing-wise, 8 supported bearings	3.40%	3.61%
Matched augmentation	Bearing-wise, same 8 supported bearings	1.32%	1.81%

When raw and K3 are both trained on clean windows plus the same noisy training realization, speed-wise raw reaches 63.83% and augmented K3 reaches 55.89%; the paired K3-minus-raw effect is -7.94 pp (unit-only decision interval [-12.61, -3.44]; crossed robustness interval [-12.72, -3.28]), with seven of ten speeds negative. At the bearing level, the corresponding pooled out-of-fold balanced accuracies are 1.32% and 1.81%, with a mean difference of 0.49 pp (unit-only interval [-0.62, 1.53]; crossed interval [-0.83, 1.67]). The clean-trained K3 advantage over a vulnerable raw baseline therefore does not survive matched noise augmentation.

The sign change is not an artifact of exact training/test noise matching. When both representations are trained at 0 dB and tested at -3 dB, the speed-wise effect is -7.39 pp (unit-only interval [-14.89, -0.61]); at +4 dB it is -2.61 pp ([-6.78, 1.17]). At -3 dB, the estimated effect remains negative and its interval excludes zero. At +4 dB, the point estimate is also negative, but the interval crosses zero. These mismatch conditions are robustness checks, not the primary comparator.

4.4 External evidence: pooled gain with class-specific harm

Figure 5 and Table 6 compare the PHM2012 external audit, the four blocks with saved class-level confusion matrices, and the primary matched-augmentation HUSTbearing speed comparison using the same interval audit. The PHM2012 clean-trained row is the clearest class-level failure: the mean effect is 24.51 pp with lower bound 19.20 pp, but the healthy class changes by 70.75 pp [57.52, 83.66] while the degraded class changes by -21.73 pp [-43.14, -5.39]. The degraded upper bound is below the illustrative -5 pp harm margin, so the block is **Not qualified** despite its positive pooled mean. Under matched augmentation, the PHM2012 mean effect falls to -6.29 pp [-12.17, -0.98] and

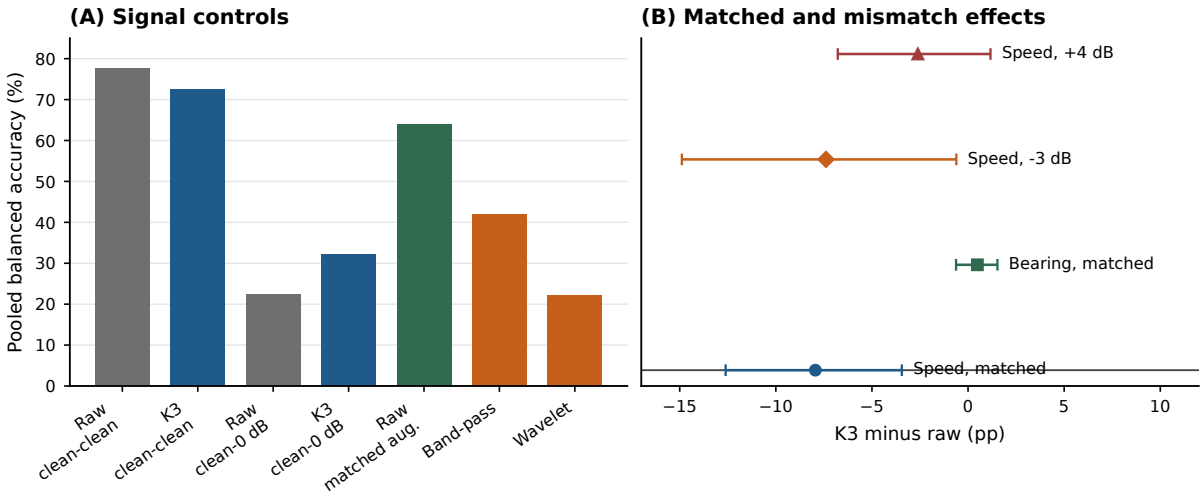


Figure 4: Matched-augmentation and signal controls on HUSTbearing. (A) Pooled balanced accuracy for clean, noise-augmented, band-pass, and wavelet conditions. (B) Paired K3-minus-raw effects with unit-only intervals.

Table 5: HUSTbearing matched-augmentation and noise-mismatch control. Unit-only intervals are decision intervals; crossed intervals in the text are robustness checks. The bearing analysis excludes the healthy bearing.

Scheme	Test condition	Units	Mean Δ	Unit-only 95% CI	Crossed 95% CI	Signs	Raw/K3 BA
Speed	matched 0 dB	10	-7.94	[-12.61, -3.44]	[-12.72, -3.28]	2/1/7	63.83/55.89
Bearing	matched 0 dB	8	+0.49	[-0.62, 1.53]	[-0.83, 1.67]	4/3/1	1.32/1.81
Speed	train 0, test -3 dB	10	-7.39	[-14.89, -0.61]	–	3/0/7	54.67/47.28
Speed	train 0, test +4 dB	10	-2.61	[-6.78, 1.17]	–	4/1/5	65.94/63.33
Bearing	train 0, test -3 dB	8	-0.83	[-2.08, 0.21]	–	2/2/4	2.92/2.08
Bearing	train 0, test +4 dB	8	+0.42	[-0.42, 1.67]	–	1/6/1	0.28/0.69

is Not qualified on the mean-effect branch.

With matched augmentation as the primary HUSTbearing baseline, the speed-level effect is -7.94 pp with unit-only interval [-12.61, -3.44]. Because the unit-only upper bound, -3.44 pp, is below $\delta = 1$ pp, the K3 decision is **Not qualified**. The crossed interval [-12.72, -3.28] is a robustness check, not the decision interval.

HSE, HUST-Hanoi, and SCA remain Insufficient because their mean-effect intervals cross zero and their class intervals do not establish that harm is within the allowed margin. Mehran-UET is Not qualified: although its mean interval crosses zero, strong inner-class damage beyond the example harm margin is present. Class intervals for HSE and HUST-Hanoi are based on only 2–5 units with positive support for each class, so they are marked fragile. No block is promoted to Qualified. The full class support, paired recall change, simultaneous interval, fragile flag, missing flag, and harm decision are reported in Appendix Table A4.

4.5 What the result means

The results support three claims in order of evidence strength. First, a pooled gain is not stable under measurement condition. The HUSTbearing clean-trained speed-level gain of 9.72 pp reverses to -7.94 pp when raw and K3 receive matched noise augmentation. The clean-trained gain is therefore a baseline-specific observation, not a preprocessing warrant.

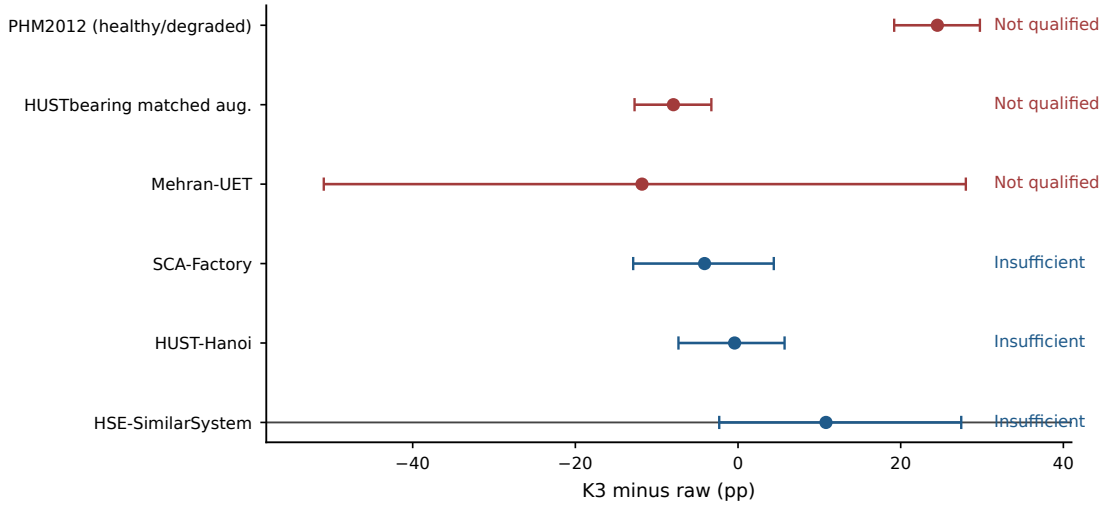


Figure 5: Evidence states for the PHM2012 run-to-failure block, the external fault-type blocks, and the matched-augmentation HUSTbearing comparison. Intervals are unit-only decision intervals.

Table 6: Worked evidence states under the illustrative margins $\delta = 1$ pp and $\eta = 5$ pp. The PHM2012 rows use a separate run-to-failure healthy/degraded endpoint with 17 independent bearings. All intervals shown are unit-only decision intervals; crossed robustness intervals are reported in the text.

Block	Units	Mean Δ (pp)	Unit-only 95% CI	Status / basis
PHM2012 clean-trained (H/D)	17	+24.51	[19.20, 29.74]	Not qualified / class harm
PHM2012 matched (H/D)	17	-6.29	[-12.17, -0.98]	Not qualified / mean effect
HSE	11	+10.81	[-2.31, 27.44]	Insufficient / mean and class inconclusive
HUST-Hanoi	33	-0.42	[-7.32, 5.72]	Insufficient / mean and class inconclusive
SCA	10	-4.11	[-12.89, 4.40]	Insufficient / mean and class inconclusive
Mehran-UET	12	-11.81	[-50.93, 28.01]	Not qualified / class harm
HUSTbearing matched aug.	10	-7.94	[-12.61, -3.44]	Not qualified / mean effect

HUSTbearing crossed robustness interval: [-12.72, -3.28]. It is not the decision interval.

Second, a positive pooled gain can conceal class-specific harm. In PHM2012, the clean-trained mean effect is 24.51 pp, but the degraded-class interval is [-43.14, -5.39] and fails the harm margin. This is stronger evidence than a generic warning about heterogeneity: the aggregate metric and the class-level decision point in opposite directions.

Third, a validation-unit holdout changes the scope of the claim. The speed-minus-bearing contrast is descriptive and is not a paired per-bearing interaction, so it cannot establish a causal interaction. It does show that a speed-level result cannot be used as a physical-bearing rule without a separate bearing-level audit. The three-state evidence audit makes these statuses explicit, but the empirical conclusion rests on the reversal and the class-specific harm, not on the labels themselves. CCMG remains an optional screen with limited incremental value.

5 Conclusion and Future Work

5.1 Conclusion

This study shows that pooled diagnostic accuracy is not sufficient to authorize VMD preprocessing. Two empirical results carry the argument. First, on the same HUSTbearing source, a clean-trained speed-level gain of 9.72 pp reverses to -7.94 pp (unit-only interval [-12.61, -3.44]; crossed robustness interval [-12.72, -3.28]) once both raw and K3 are trained with matched 0 dB augmentation. Second, in the independent PHM2012 run-to-failure audit with 17 non-fragile bearings, clean-trained K3 raises mean balanced accuracy by 24.51 pp, but the healthy class gains 70.75 pp while the degraded class loses 21.73 pp. The pooled gain therefore coexists with an entirely negative degraded-class interval, and the block is Not qualified on class-harm grounds. Matched augmentation changes the PHM2012 mean effect to -6.29 pp.

No external block receives a full Qualified label. The matched-augmentation HUSTbearing speed audit is Not qualified under the illustrative $\delta = 1$ pp margin; its bearing-level result is unresolved; HSE, HUST-Hanoi, and SCA remain Insufficient; Mehran-UET is Not qualified due to inner-class harm; PHM2012 is Not qualified under both its clean-trained and matched protocols. The three-state audit makes these evidence statuses explicit, but it is an audit template rather than a validated certification system. CCMG remains an optional prefilter with small incremental value.

The paper does not establish a universal K3 rule, a benefit predictor, or a deployment recommendation. It establishes that pooled accuracy can misauthorize preprocessing when measurement conditions, validation units, or class-specific effects are not included in the decision. Raw remains the conservative default.

5.2 Future work

Matched noise augmentation should be repeated across the remaining external blocks and across multiple noise realizations in PHM2012. PHM2012 uses a different binary healthy/degraded endpoint and a single fixed noise realization, so its result should not be read as a five-class replication or a field validation. HSE, HUST-Hanoi, and SCA remain Insufficient under their clean-trained protocols, whereas Mehran-UET is Not qualified under the illustrative class-damage margin.

Class-complete validation units are also needed. HSE and HUST-Hanoi have positive class support in only 2–5 units, which makes their class-level intervals fragile and prevents a full Qualified decision. A prospective target-unit study should pre-specify power or precision before fixing the number of validation units; the synthetic boundary case in Appendix Figure A4 shows that ten units yield only about 15% Qualified under borderline benefit, so ten units are not a default minimum sample size.

The clean-baseline features are signal-quality indicators, not calibrated sensor uncertainties. A measurement-chain study that records sensor type, mounting, sampling rate, and maintenance state would make the CCMG threshold more transportable. A fault-physics mechanism analysis would require trusted shaft speed and bearing geometry, which the present evidence does not provide. Until such evidence exists, the three-state audit should be read as a decision aid for declared validation units, not as a universal deployment rule.

Data and Code Availability

The analysis code, frozen protocols, generated tables, figures, and result files accompany the submission. Source datasets retain their original licenses and access conditions. The analysis artifact is available from the corresponding author upon reasonable request.

Ethics Declaration

Not applicable. This study uses previously collected bearing vibration datasets and does not involve human participants, animals, or personally identifiable information.

CRediT Author Contributions

The authors jointly designed the study, curated the datasets, implemented the analysis, interpreted the results, and wrote the manuscript.

Conflict of Interest

The authors declare no conflict of interest.

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AI Disclosure

AI-assisted tools were used for analysis scripting, manuscript drafting, and internal consistency checks. All numerical results were generated from the documented analysis artifacts. The authors have verified the numerical outputs against the artifacts.

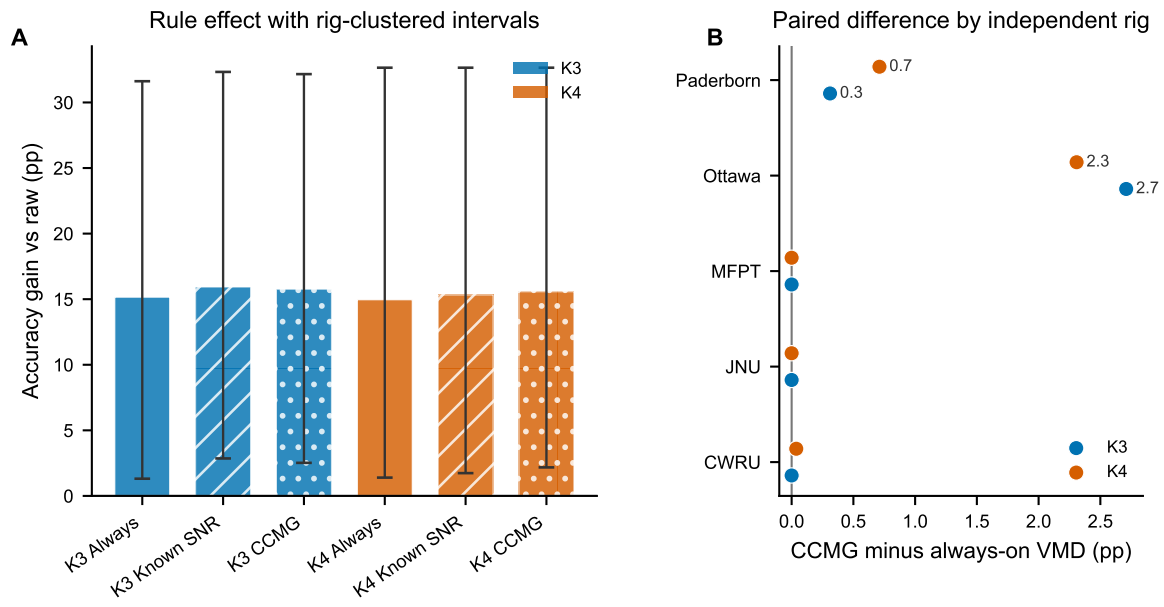


Figure A1: CCMG ablation. The optional gate changes only a small number of comparisons and does not predict downstream K3 benefit.

A Additional controls

A.1 CCMG ablation

Figure A1 shows the ablation result.

The CCMG screen avoided 2 of 86 comparisons and changed the evidence-block-equalized selected effect by 0.30 pp. Its perfect clean/noise separation in the controlled grid does not establish field-noise recognition or diagnostic-benefit prediction.

A.2 Decomposition-family comparison

Figure A2 reports the protocol-conditional comparison.

A.3 Nested mode selection

Figure A3 and Table A2 summarize the nested selection and run-level audit.

B Operating characteristics and field actions

Figure A4 reports the synthetic operating characteristics, and Table A3 maps each state to a field action.

The boundary-benefit scenario yields Qualified rates of roughly 15–25% and mostly Insufficient. The boundary-harm scenario produces roughly 3–4% false Qualified decisions. These results depend on the stated simulation assumptions and do not establish coverage under the dependence structure of the observed datasets.

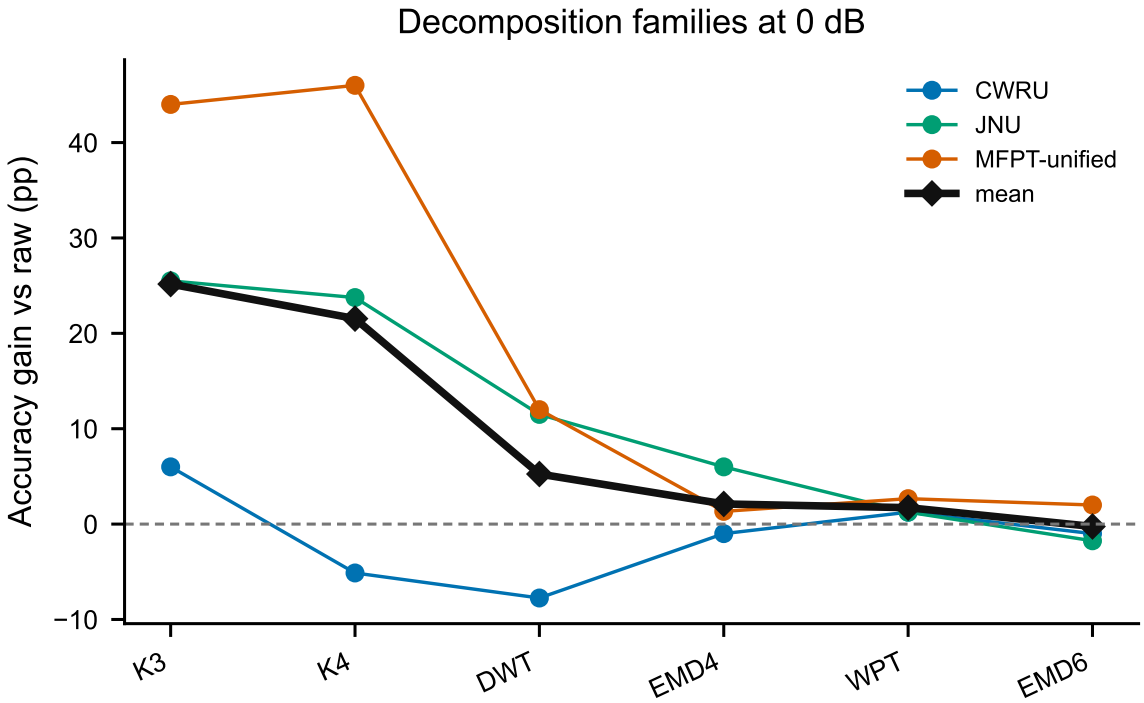


Figure A2: Protocol-conditional decomposition-family comparison. The comparison is relative to the vulnerable clean-trained raw baseline and does not establish a general VMD advantage.

C Class-level evidence audit

The table exposes the evidence behind each worked evidence state. The decision column is evaluated against the same illustrative harm margin used in the main text. Missing or inconclusive class evidence is not converted into a pass. Mehran-UET has strong evidence of inner-class harm and no comparable evidence of outer-class harm, whereas HSE and HUST-Hanoi have several inconclusive classes because their bearing-level folds do not contain every class in every validation unit.

D Benefit-predictor audit

Figure A5 reports the evidence-block benefit-predictor audit.

The four-feature logistic model reached AUROC 0.651 and the ridge model reached AUROC 0.514. The blind-noise feature alone reached AUROC 0.687. The ridge point estimate is 8.64 pp, slightly above the 7.79 pp all-K3 value in the policy audit, but its near-chance discrimination and mean absolute error of 22.4 pp do not support a reliable selector interpretation.

E Reproducibility and limitations

This is a secondary analysis of completed measurements and classifier runs. All severe-noise results use additive white Gaussian noise, not a field-noise campaign. The discovery synthesis combines completed classifier runs from several protocols. The external audits share layer widths, optimizer, learning rate, batch size, and training budget, but K3 has three input channels and therefore 576

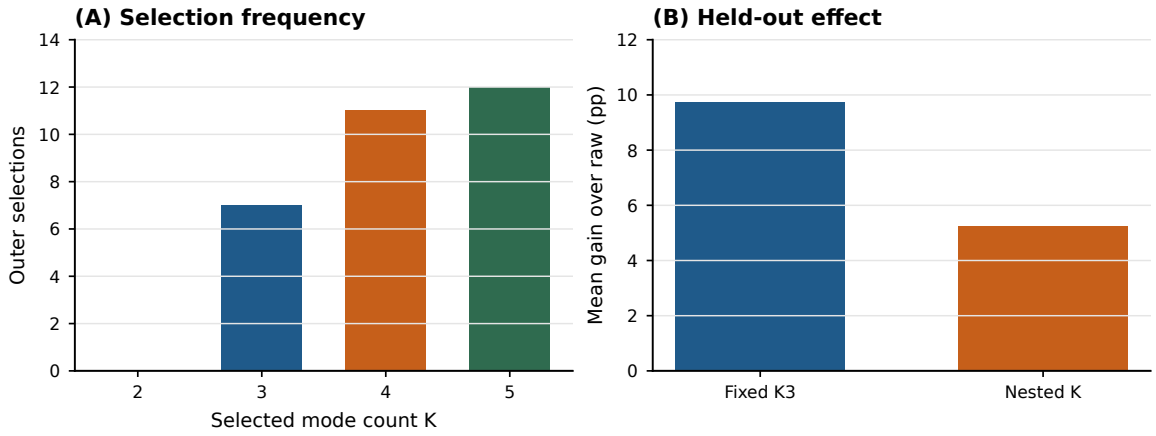


Figure A3: Nested K selection on HUSTbearing. Inner validation selects K3 in only 7 of 30 outer-fold/seed selections; fixed K3 is a controlled comparator, not a selected policy.

Table A1: raw3 capacity control and evidence-block summary. The control repeats the one-channel raw window three times and is exploratory.

Block	Unit	n	K3–raw (pp)	K3–raw3 (pp)
CWRU	condition/file fold	1	+6.00	–
JNU	condition fold	1	+25.50	–
MFPT	rig protocol	1	+44.00	–
Ottawa	condition fold	4	+0.59	–
Paderborn	protocol fold	3	-0.31	–
HUSTbearing	motor speed	10	+7.61	+7.22
HSE	physical bearing	11	+10.81	+10.45
HUST-Hanoi	physical bearing	33	-0.42	-0.34
SCA	factory case	10	-4.11	-4.46
Mehran-UET	fault-size group	12	-11.81	-12.04

additional first-layer parameters. The comparison is not a pure VMD-representation effect isolated from every architecture difference.

HUST-Hanoi and HSE each contain a finite number of physical bearings, and several operating conditions belong to the same bearing. The exact mode count remains protocol dependent. K3 was fixed before the external audits but was not selected on an independent validation partition within the discovery data. The other external blocks still use fixed test realizations and clean-trained raw. The crossed bootstrap treats the finite seed and noise levels as resampling factors, so its interval is a sensitivity interval rather than a large-sample guarantee. The synthetic simulation is not an empirical coverage calibration.

All numerical tables are generated from saved CSV or JSON artifacts. The analysis manifest records hashes for source summaries, scripts, figures, and reports. No result was filtered by outcome direction. The neutral HUST-Hanoi result, the negative HUSTbearing fold, the negative SCA and Mehran-UET means, and the negative HSE bearings are retained.

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Table A2: Run-level audit summary. Repeat counts describe repeated experimental combinations, not additional independent validation units.

Block	Validation unit	Units	Seeds	Noise realizations	Protocol role
HUSTbearing, speed	motor speed	10	3	5	primary split audit
HUSTbearing, bearing	physical bearing	8	3	5	same-source split
HSE	physical bearing	11	2	1	external bearing audit
HUST-Hanoi	physical bearing	33	2	1	external bearing audit
SCA	factory case	10	2	1	industrial audit
Mehran-UET	fault-size group	12	2	1	industrial audit
PHM2012 clean	run-to-failure bearing	17	3	1	locked external audit
PHM2012 matched	run-to-failure bearing	17	3	1	locked external audit

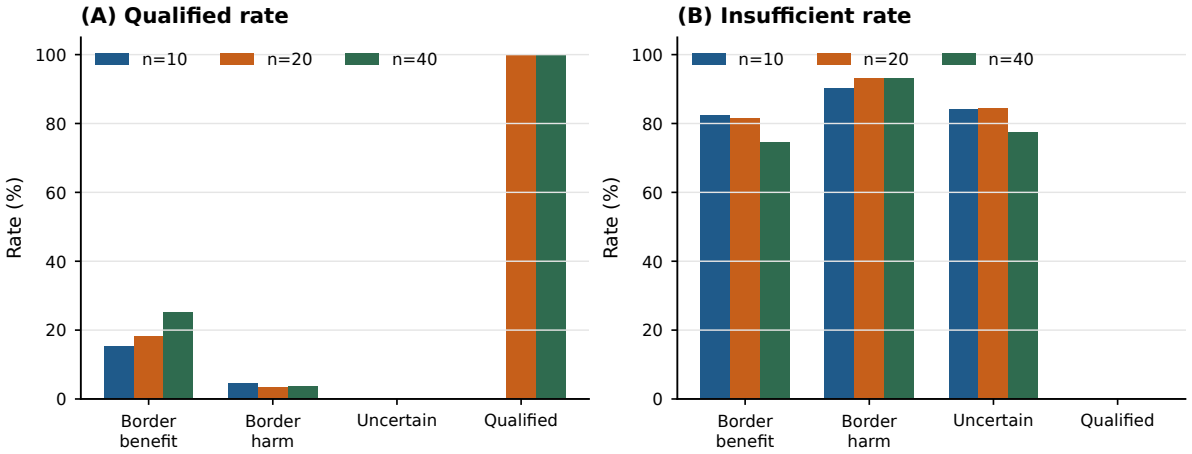


Figure A4: Synthetic operating characteristics of the three-state rule under independent validation units and Gaussian class-level changes. The simulation is not an empirical coverage calibration.

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Table A3: Field actions for the three authorization states.

State	Immediate action	Required follow-up
Qualified	Enable the candidate only for the declared validation unit and operating envelope; log the version and margins.	Monitor drift and re-qualify after any change in sensor, mounting, speed/load, bearing type, fault mix, or baseline condition.
Not qualified	Keep raw as the production default; do not enable the candidate for that unit.	Reopen only with a new prespecified comparator, unit, or measurement condition; do not reinterpret the same test outcome as a qualified result.
Insufficient	Keep raw or use a controlled engineering pilot without a production benefit claim.	Add independent units, missing class support, or the missing class-level interval; then repeat Stage 2. Do not resolve insufficiency by weakening δ, η, or the confidence level.

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Table A4: Class-level evidence audit for the worked examples, including the separate PHM2012 healthy/degraded run-to-failure endpoint. Intervals are Bonferroni-adjusted over classes; support counts reflect validation units in which the class occurs. Fragile marks intervals supported by two to five independent units. Missing marks classes without a valid interval. The harm margin is $\eta = 5$ pp. A class interval marked fragile does not by itself license a Qualified decision.

Block	Class	Support	Mean	Simultaneous interval	Fragile	Missing	Harm decision
PHM2012 clean-trained	healthy	17	+70.75	[57.52, 83.66]	no	no	within
PHM2012 clean-trained	degraded	17	-21.73	[-43.14, -5.39]	no	no	harm
HSE-SimilarSystem	inner	3	+0.56	[0.00, 1.67]	yes	no	within
HSE-SimilarSystem	normal	3	+36.67	[5.42, 79.17]	yes	no	within
HSE-SimilarSystem	outer	2	+9.17	[0.00, 18.33]	yes	no	within
HSE-SimilarSystem	rolling_element	3	-3.68	[-24.58, 20.62]	yes	no	inconclusive
HUST-Hanoi	inner	5	+0.00	[0.00, 0.00]	yes	no	within
HUST-Hanoi	inner+outer	5	+3.33	[-8.33, 20.00]	yes	no	inconclusive
HUST-Hanoi	inner+rolling_element	4	+0.69	[0.00, 2.08]	yes	no	within
HUST-Hanoi	normal	5	-31.67	[-62.22, -2.22]	yes	no	inconclusive
HUST-Hanoi	outer	5	+7.78	[0.00, 20.00]	yes	no	within
HUST-Hanoi	outer+rolling_element	5	+12.22	[1.11, 36.11]	yes	no	within
HUST-Hanoi	rolling_element	4	+6.25	[0.00, 19.44]	yes	no	within
SCA-Factory	normal	10	+0.83	[-25.42, 26.00]	no	no	inconclusive
SCA-Factory	fault	10	-9.06	[-38.59, 19.76]	no	no	inconclusive
Mehran-UET	inner	6	-74.07	[-95.83, -50.00]	no	no	harm
Mehran-UET	outer	6	+50.46	[14.81, 82.41]	no	no	within

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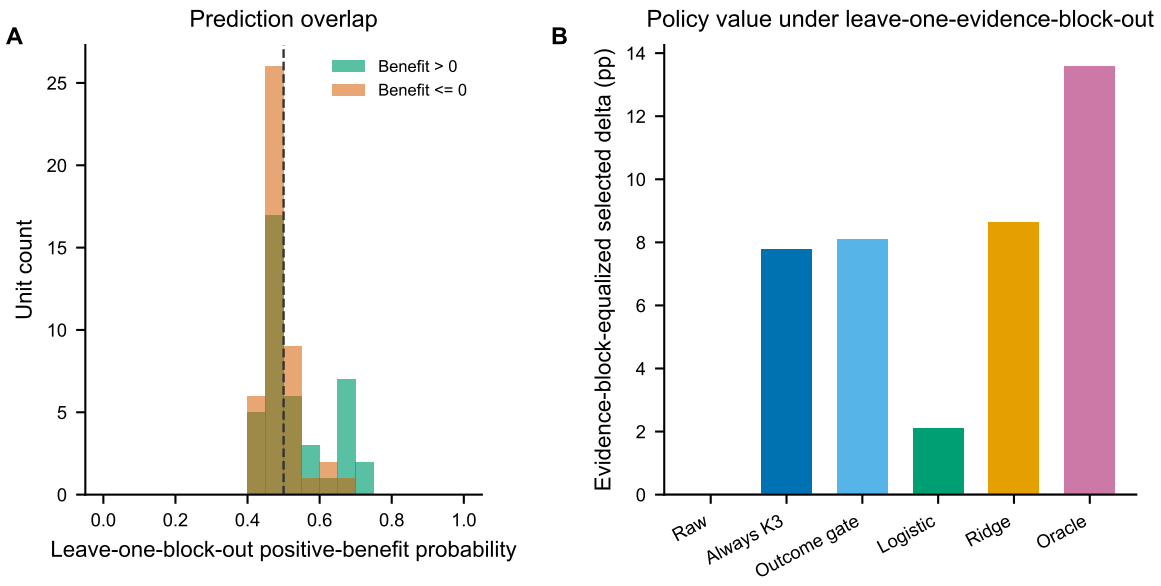


Figure A5: Evidence-block benefit-predictor audit. The four-feature models do not support a reliable cross-block selection rule.